

Algorithms: Quiz 2

19 Mar 2026

Total Marks: 35

Time: 2 hrs 15 min

1. Find an optimal Huffman code for the following set of letter frequencies: [4 marks]

Letter	a	b	c	d	e	f	g
Frequency	2	2	3	4	6	8	12

2. A directed acyclic graph G is given by the following adjacency lists of out-neighbors:

v	A	B	C	D	E	F	G	H
Out(v)	B,D	C,D,E	E,H	$\emptyset$	D	C,H	A,D	E

Illustrate how to find a topological ordering of the above graph.

[4 marks]

3. A *shuffle* of two strings X and Y is obtained by interleaving the characters into a new string, keeping the characters of X and Y in the same order.

For example, 'bananaanas' is a shuffle of 'banana' and 'anas' in many ways: BANANAanas, BANanaANAs, BanANaAnaNs (the capital letters belong to the first string). The string 'dyprongarmammicing' is a shuffle of 'dynamic' and 'programming': DYproNgArMammICing.

Given three strings $A[1..m]$, $B[1..n]$ and $C[1..m+n]$, describe and analyze an algorithm to determine whether C is a shuffle of A and B .

[5 marks]

4. Suppose we have n skiers with heights given in an array $P[1, \dots, n]$, and n skis with heights given in an array $S[1, \dots, n]$. Describe an efficient algorithm to assign a ski to each skier, so that the average difference between the height of a skier and her assigned ski is as small as possible. The algorithm should compute a permutation σ such that the expression

$$\frac{1}{n} \sum_{i=1}^n |P[i] - S[\sigma(i)]|$$

is as small as possible. Argue about the correctness of your algorithm as well. [5 marks]

5. The game of ladders-without-snakes uses a board which is a $n \times n$ grid of squares, numbered from 1 to n^2 , starting in the left bottom corner and proceeding from bottom to top; certain pairs of squares, in different rows, are connected by a ladder. Each square can be the endpoint of at most one ladder. See example below.

100	99	98	97	96	95	94	93	92	91
81	82	83	84	85	86	87	88	89	90
80	79	78	77	76	75	74	73	72	71
61	62	63	64	65	66	67	68	69	70
60	59	58	57	56	55	54	53	52	51
41	42	43	44	45	46	47	48	49	50
40	39	38	37	36	35	34	33	32	31
21	22	23	24	25	26	27	28	29	30
20	19	18	17	16	15	14	13	12	11
1	2	3	4	5	6	7	8	9	10

You start with a token in square 1 and in each move, you can advance your token at most k steps, where k is a constant fixed initially. If your token lands at the foot of a ladder, in zero moves, your token immediately moves forward to the top of the ladder.

Describe an efficient algorithm to find the shortest sequence of moves required to reach the last square, given the list of ladders and board size as input. [6 marks]

8. There is a building with some rooms and there are n activities that are planned to be held in these rooms. Only one activity can be held at a time in one room and it must be completed once started. The start and finish time of each activities is given as input. Suppose that we wish to schedule these activities while minimizing the number of rooms being used.

Consider each of the following two greedy algorithms and decide which one is correct, i.e. gives the optimal solution (with justification), and which one is incorrect (with a counterexample).

Algorithm A: Process the activities in increasing order of finish times, and assign each activity the first available room.

Algorithm B: Process the activities in increasing order of start times, and assign each activity the first available room.

For example, if there are 3 activities whose intervals are $[1,4]$, $[2,3]$, $[5,6]$, then Algorithm A would first assign room 1 to $[2,3]$, then room 2 to $[1,4]$, and then room 1 to $[5,6]$.

Algorithm B would first assign room 1 to $[1,4]$, then room 2 to $[2,3]$, and then room 1 to $[5,6]$.

[6 marks]

7. There is a set A of companies and a set B of candidates who are looking for jobs. Each company in A has shortlisted some candidates from B whom they are willing to offer a job. However, each company can make a maximum of two offers as they have only 2 vacancies each.

The input to the problem is a bipartite graph $G = (V, E)$ with $V = A \cup B$ and edges between $u \in A$ and $v \in B$ indicating that company u is willing to offer candidate v a job.

Design an algorithm to maximize the number of candidates who receive jobs.

[5 marks]